



Database Management Systems

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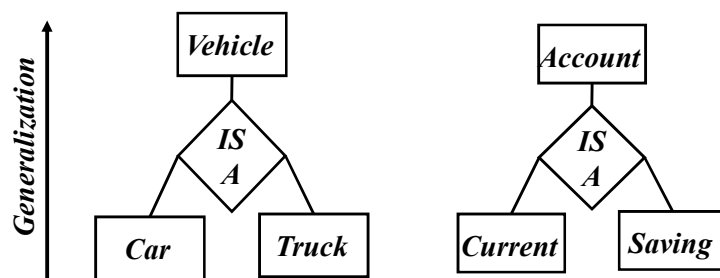


Relational Database Management Systems

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Generalization

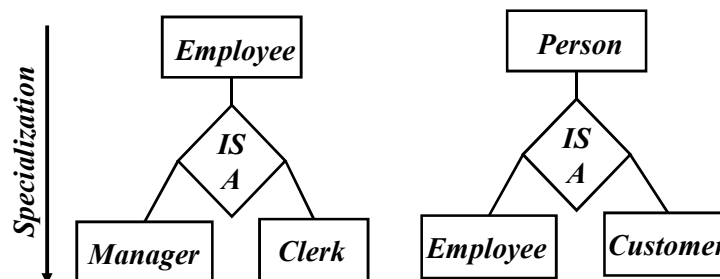
- ❑ A bottom up approach in database design to **combine** two or more low level entities that share same attributes into a generalized higher level entity.
- ❑ Focus on similarities.
- ❑ Generalization is conceptual synthesis



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Specialization

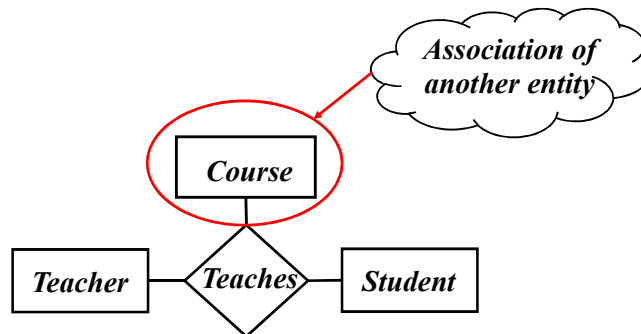
- ❑ A top down approach in database design to **divide** a higher level entity into two or more low level entities based on specific attributes or roles.
- ❑ Focus on differential properties.
- ❑ Specialization is conceptual refinement.



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Aggregation

- ❑ Aggregation is an abstraction concept that represents a relationship as an entity.
- ❑ It is used when a relationship itself has attributes or when a relationship involves other relationships.
- ❑ More organized representation of complex relationship.



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Anamolies in Relational Database

- ❑ Problems that arise when the database is poorly designed, particularly when **normalization** is not applied properly.
- ❑ The anomalies can lead to **inconsistencies, redundancy, and inefficiency** in the database.
 - ❑ *Insertion anomalies*
 - ❑ *Modification anomalies*
 - ❑ *Deletion anomalies*

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Insertion Anomaly



- Occurs when the design of a table makes it difficult to add new data without leaving certain fields empty or introducing redundant data.

❑ **Cause:** Improper separation of entities, leading to dependency issues.

StuID	StuName	CourseID	CouName
101	Vineet	CS101	DBMS
102	Ayush	CS101	DBMS
103	Surya	CS102	Micro
104	Ashish	CS102	Micro
105	Anshu	CS103	COA
106	Sumit	CS104	C++

- Can we add a new course details in the table?
- Only if some student enroll in that course otherwise we cannot.

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Deletion Anomaly



- Occurs when the deletion of a record inadvertently removes valuable information from the database.

❑ **Cause:** Storing unrelated data in the same table

StuID	StuName	CourseID	CouName
101	Vineet	CS101	DBMS
102	Ayush	CS101	DBMS
103	Surya	CS102	Micro
104	Ashish	CS102	Micro
105	Anshu	CS103	COA
106	Sumit	CS104	C++

- What if the only enrolled student drops out and their record is deleted?
- The course information will be lost.

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Update Anomaly



- ❑ Occurs when redundant data leads to inconsistencies after an update operation.
 - ❑ **Cause:** Redundancy in data storage.

StuID	StuName	CourseID	CouName
101	Vineet	CS101	DBMS
102	Ayush	CS101	DBMS
103	Surya	CS102	Micro
104	Ashish	CS102	Micro
105	Anshu	CS103	COA
106	Sumit	CS104	C++

- What if the course name for CS101 changes from **DBMS** → **RDBMS**?
- All rows needs to be updated, failing to update will lead to inconsistent data.

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How to Avoid Anomalies?



- ❑ **Normalization:** Break down tables into smaller, related tables to ensure data is stored logically and without redundancy.
 - ❑ Refinement process (e.g. one table should have single or concise data).
 - ❑ How → by removing dependencies (1NF, 2NF, 3NF, BCNF).
- ❑ **Use Primary and Foreign Keys:** Ensure proper relationships between tables using keys.
- ❑ **Separate Concerns:** Store distinct types of data in separate tables (e.g. separate Student and Course entities).

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Avoid Anomalies: An Example

<u>StuID</u>	StuName	CourseID	CouName
101	Vineet	CS101	DBMS
102	Ayush	CS101	DBMS
103	Surya	CS102	Micro
104	Ashish	CS102	Micro
105	Anshu	CS103	COA
106	Sumit	CS104	C++

PK

- Can we insert new course details?
- Does deletion of a student record impact course details?
- Can we update course details efficiently?

<u>StuID</u>	StuName	CourseID
101	Vineet	CS101
102	Ayush	CS101
103	Surya	CS102
104	Ashish	CS102
105	Anshu	CS103
106	Sumit	CS104

<u>CourseID</u>	CouName
CS101	DBMS
CS102	Micro
CS103	COA
CS104	C++

FK

PK

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Functional Dependencies

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Dependency

Tata Motors

➔



- If it's a Tata Motors car then it must be a Nexon

Tata Motors

➔



- If the car is Nexon, then it must be from Tata Motors.

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Dependency

Branch_Code

➔

HoD Name

- If we are given branch code, can we tell who is HoD for that branch?

HoD_Name

➔

Branch_Code

- If we are given name of the HoD, can we tell what is the branch code of the HoD?

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Functional Dependency



- ❑ It describes the relationship between attributes in a relation.
- ❑ In a relationship R , functional dependency, denoted by $X \rightarrow Y$, between two sets of attributes X and Y that are subsets of R specifies a constraint on the possible tuples that can form a relation state r of R .
- ❑ The constraint is that, for any two tuples $T1$ and $T2$ in r that:
 - ❑ have $T1[X] = T2[X]$,
 - ❑ they must also have $T1[Y] = T2[Y]$.

X	Y	Z
1	4	2
1	4	3
2	6	3
3	2	2
2	6	3

- Can we say $X \rightarrow Y$?
- Can we say $Y \rightarrow Z$?
- Can we say $XY \rightarrow Z$?

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Armstrong's Axioms



- ❑ Armstrong's axioms (base standard rules) are a set of inference rules used to derive all possible functional dependencies in a relational database.
 - ❑ **Reflexivity:** if X is a set of attributes and Y is subset of X , then $X \rightarrow Y$ holds.
 - ❑ **Augmentation:** If $X \rightarrow Y$ holds and Z is a set of attributes, then $ZX \rightarrow ZY$ or $XZ \rightarrow YZ$ holds.
 - ❑ **Transitivity:** If $X \rightarrow Y$ holds and $Y \rightarrow Z$ holds, then $X \rightarrow Z$ holds.

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RAKUMC

Secondary Rules



- ☐ **Union:** If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$ holds.
 - ☐ **Decomposition:** If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$ holds.
 - ☐ **Pseudo Transitivity:** If $X \rightarrow Y$ and $YW \rightarrow Z$, then $XW \rightarrow Z$ holds.
 - ☐ **Composition:** If $X \rightarrow Y$, then $Z \rightarrow W$, then $XZ \rightarrow YW$ holds.
- ☐ **Note:** Armstrong's axioms are *sound* and *complete*. These inference axioms can be derived from Armstrong's axioms.

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Trivial and Non-Trivial Dependency



- ☐ **Trivial:** A functional dependency $X \rightarrow Y$ is trivial if Y is subset of X .
- ☐ **Non-Trivial:** A functional dependency $X \rightarrow Y$ is non-trivial if Y is not a subset of X .

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Proving Union



- **Given:** If $X \rightarrow Y$ and $X \rightarrow Z$
- $XX \rightarrow XY$; (augmentation rule: $X \rightarrow Y$).
 - $X \rightarrow XY$; (since: $XX \rightarrow X$).
 - $XY \rightarrow YZ$; (augmentation rule: $X \rightarrow Y$ and $X \rightarrow Z$).
 - $X \rightarrow YZ$; (transitivity rule: $X \rightarrow XY$, $XY \rightarrow YZ$).

□ Example:

StuID	Name	Course	Year
101	Vineet	DBMS	2025
102	Ayush	PHY	2024

$\text{StuID} \rightarrow \text{Name}$
 $\text{StuID} \rightarrow \text{Course}$

$\text{StuID} \rightarrow \text{Name, Course}$

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Proving Pseudo Transitivity



- **Given:** If $X \rightarrow Y$ and $YW \rightarrow Z$
- $XW \rightarrow YW$; (augmentation rule: $X \rightarrow Y$).
 - $XW \rightarrow Z$; (transitivity rule: $XW \rightarrow YW$ and $YW \rightarrow Z$).

StuID	CouID	Instructor	Grade
101	CS101	DNY	A
102	CS102	DPM	Ex

$\text{StuID} \rightarrow \text{CouID}$
 $\text{CouID, Instructor} \rightarrow \text{Grade}$

$\text{StuID, Instructor} \rightarrow \text{Grade}$

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Proving Decomposition



□ **Given:** If $X \rightarrow YZ$

- $YZ \rightarrow Y$; (reflexivity rule: because Y is subset of YZ).
- $YZ \rightarrow Z$; (reflexivity rule: because Z is subset of YZ).
- $X \rightarrow Y$; (transitivity rule: $X \rightarrow YZ$, and $YZ \rightarrow Y$).
- $X \rightarrow Z$; (transitivity rule: $X \rightarrow YZ$, and $YZ \rightarrow Z$).

ProID	Name	Price
101	Laptop	50000
102	Phone	25000

$\text{ProID} \rightarrow \text{Name, Price}$  $\text{ProID} \rightarrow \text{Name}$
 $\text{ProID} \rightarrow \text{Price}$

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Proving Composition



□ **Given:** $X \rightarrow Y, W \rightarrow Z$

- $XW \rightarrow YW$; (augmentation rule: add W both side).
- $YW \rightarrow YZ$; (augmentation rule: add Y both side).
- $XW \rightarrow YZ$; (transitivity rule: $XW \rightarrow YW$ and $YW \rightarrow YZ$).

StuID	CouID	Name	Grade
101	CS101	Vineet	A
102	CS102	Ayush	Ex

$\text{StuID} \rightarrow \text{Name}$  $\text{StuID, CouID} \rightarrow \text{Name, Grade}$
 $\text{CouID} \rightarrow \text{Course}$

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Summary of Rules			
Rule	Statement	Example	Axiom Used
Reflexivity	If $Y \subseteq X$ then $X \rightarrow Y$	$\{A, B, C\} \rightarrow \{A\}$	Fundamental
Augmentation	If $X \rightarrow Y$ then $XZ \rightarrow YZ$	$\{A\} \rightarrow \{B\}$ $\Rightarrow \{A, C\} \rightarrow \{B, C\}$	Fundamental
Transitivity	If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$	$\{A\} \rightarrow \{B\}, \{B\} \rightarrow \{C\}$ $\Rightarrow \{A\} \rightarrow \{C\}$	Fundamental
Union	If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow YZ$.	$\{A\} \rightarrow \{B\}, \{A\} \rightarrow \{C\}$ $\Rightarrow \{A\} \rightarrow \{B, C\}$	Derived (Aug + Trans)
Decomposition	If $X \rightarrow YZ$ then $X \rightarrow Y$ and $X \rightarrow Z$.	$\{A\} \rightarrow \{B, C\}$ $\Rightarrow \{A\} \rightarrow \{B\}, \{A\} \rightarrow \{C\}$	Derived (Ref + Trans)
Pseudotransitivity	If $X \rightarrow Y$ and $YW \rightarrow Z$ then $XW \rightarrow Z$	$\{A\} \rightarrow \{B\}, \{B, C\} \rightarrow \{D\}$ $\Rightarrow \{A, C\} \rightarrow \{D\}$	Derived (Aug + Trans)
Composition	If $X \rightarrow Y$ and $W \rightarrow Z$, then $XW \rightarrow YZ$	$\{A\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}$ $\Rightarrow \{A, C\} \rightarrow \{B, D\}$	Derived (Aug + Trans)

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Functional Dependencies and Closure	
❑ The set of all functional dependencies (FDs) implied by a given set F of FDs is called the closure of F, and denoted as F^+ ❑ Armstrong Axioms can be applied repeatedly to infer all FDs implied by a set F of FDs. ❑ Sound: The axioms generate ONLY FDs in F^+ when applied to a given set of FDs F. ❑ Complete: The axioms, when repeatedly applied to a given set of FDs F, will generate ALL FDs in F^+	

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Implication of Functional Dependencies

- Let functional dependency set $FD = \{AB \rightarrow CD, B \rightarrow DE, C \rightarrow F, E \rightarrow G, A \rightarrow B\}$. Use Armstrong's axioms to derive that $A \rightarrow FG$ is logically implied by FD .

Step#	Inference	Justification
1	$A \rightarrow B$	Given
2	$A \rightarrow AB$	Augmentation of A on step 1
3	$AB \rightarrow CD$	Given
4	$A \rightarrow CD$	Transitivity on steps 2,3
5	$B \rightarrow DE$	Given
6	$A \rightarrow DE$	Transitivity on steps 1,5
7	$A \rightarrow ACD$	Augmentation of A on step 4
8	$ACD \rightarrow CDE$	Augmentation of C,D on step 6
9	$A \rightarrow CDE$	Transitivity on steps 7,8
10	$A \rightarrow CE$	Trivial dependency from step 9
11	$C \rightarrow F$	Given
12	$CE \rightarrow EF$	Augmentation of E on step 11
13	$E \rightarrow G$	Given
14	$FE \rightarrow FG$	Augmentation of F on step 13
15	$CE \rightarrow FG$	Transitivity on steps 12,14
16	$A \rightarrow FG$	Transitivity on steps 10,15

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Attribute Closure

- For a given FD set, Attribute closure of an attribute set can be defined as set of attributes which can be functionally determined by using axioms and FD set.
- Closure of A is denoted by $\{F\}^+$ or $(A)^+$
- Closure of $AB = (AB)^+ = \{A^+ \cup B^+ \cup (\text{Any FD in } F \text{ where } AB \text{ is the determinant})\}$

$R(A, B, C, D)$	$R(A, B, C, D, E, F, G)$
$A \rightarrow B$	$A \rightarrow B$
$A \rightarrow C$	$BC \rightarrow DE$
$AB \rightarrow D$	$AEG \rightarrow G$
$A^+ = ?$	$AC^+ = ?$
$A \rightarrow A \rightarrow AB \rightarrow ABCD$	$AC \rightarrow ABC \rightarrow ABCDE$
$\rightarrow ABCD$	

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Computing Closure of FD set



- ❑ **Step 1:** Compute S, which is the set all attributes in the FD set
- ❑ **Step 2:** Compute P(S), which is the power set of S except null element
- ❑ **Step 3:** Compute closure of each element of P(S)
- ❑ **Step 4:** If the closure of an element of P(S) is of the form $\{X\}^+ = \{Y\}$, then $(2^{|Y|} - 1)$ number of FDs will be found from this. The FDs will be of the form $X \rightarrow Z$ where Z is any element in P(Y) (power set of Y) except null.

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Computing Closure of FD set



Systematically computing Closure of an FD set:

Find closure of $F = \{A \rightarrow B, A \rightarrow C, B \rightarrow C\}$

Attribute Closure	Derived FDs
$A^+ = \{ABC\}$	$A \rightarrow A, A \rightarrow B, A \rightarrow C, A \rightarrow AB, A \rightarrow BC, A \rightarrow AC, A \rightarrow ABC$
$B^+ = \{BC\}$	$B \rightarrow B, B \rightarrow C, B \rightarrow BC$
$C^+ = \{C\}$	$C \rightarrow C$
$(AB)^+ = \{ABC\}$	$AB \rightarrow A, AB \rightarrow B, AB \rightarrow C, AB \rightarrow AB, AB \rightarrow BC, AB \rightarrow AC, AB \rightarrow ABC$
$(BC)^+ = \{BC\}$	$BC \rightarrow B, BC \rightarrow C, BC \rightarrow BC$
$(AC)^+ = \{ABC\}$	$AC \rightarrow A, AC \rightarrow B, AC \rightarrow C, AC \rightarrow AB, AC \rightarrow BC, AC \rightarrow AC, AC \rightarrow ABC$
$(ABC)^+ = \{ABC\}$	$ABC \rightarrow A, ABC \rightarrow B, ABC \rightarrow C, ABC \rightarrow AB, ABC \rightarrow BC, ABC \rightarrow AC, ABC \rightarrow ABC$
Closure of F	All the FDs above in this column

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Computing Closure of FD set



Q. In a schema with attributes A, B, C, D, E, following set of functional dependencies are given: $A \rightarrow B$; $A \rightarrow C$; $CD \rightarrow E$; $B \rightarrow D$; $E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set?

- (a) $CD \rightarrow AC$
- (b) $BD \rightarrow CD$
- (c) $BC \rightarrow CD$
- (d) $AC \rightarrow BC$

[GATE2005]

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Computing Closure of FD set



Q. In a schema with attributes A, B, C, D, E, following set of functional dependencies are given: $A \rightarrow B$; $A \rightarrow C$; $CD \rightarrow E$; $B \rightarrow D$; $E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set?

- (a) $CD \rightarrow AC$
- (b) $BD \rightarrow CD$**
- (c) $BC \rightarrow CD$
- (d) $AC \rightarrow BC$

[GATE2005]

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Extraneous Attribute



- ❑ For a given FD set F , an attribute A is extraneous in $X \rightarrow Y$ if A can be removed from the left side or right side of $X \rightarrow Y$ without altering the closure of F .
- ❑ Let $G = \{A \rightarrow BC, B \rightarrow C, AB \rightarrow D\}$
- ❑ Attribute C is extraneous in the right side of $A \rightarrow BC$
 - i.e., $G' = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$ has same closure as G .
- ❑ Attribute B is extraneous in the left side of $AB \rightarrow D$
 - i.e., $G'' = \{A \rightarrow BC, B \rightarrow C, A \rightarrow D\}$ has same closure as G .

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The Satisfies Algorithm



- ❑ Used to determine if a relation R satisfies or doesn't satisfy a given FD: $A \rightarrow B$
- Input:** Relation R and an FD: $A \rightarrow B$
- Output:** TRUE if R satisfies $A \rightarrow B$, otherwise FALSE
- **Step 1:** Sort the tuples of the relation R on the attribute(s) A (determinant) so that tuples with equal values under A are next to each other
- **Step 2:** Check that tuples with equal values under A also have equal values under attribute(s) B
- **Step 3:** If any two tuples of R have equal values under A but different values under attribute(s) B , output of the algorithm is FALSE
- **Step 4:** If every two tuples of R having equal values under A also have same values under attribute(s) B , output of the algorithm is TRUE

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The Satisfies Algorithm: Example



X	Y	Z
1	4	2
1	4	3
2	6	3
3	2	2
2	6	3

- Can we say $X \rightarrow Y$?
- Can we say $Y \rightarrow Z$?
- Can we say $XY \rightarrow Z$?

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The Satisfies Algorithm: Example



Consider the relation TABLE_PURCHASE_DETAIL(Customer_ID, Store_ID, Purchase_Location)

TABLE_PURCHASE_DETAIL

CustomerID	Store ID	Purchase Location
1	1	Los Angeles
1	3	San Francisco
2	1	Los Angeles
3	2	New York
4	3	San Francisco

Check if the following functional dependencies are satisfied in the above relation:

- Q1. Customer_ID $\rightarrow$ Purchase_Location
- Q2. Store_ID $\rightarrow$ Purchase_Location
- Q3. {Customer_ID, Store_ID} $\rightarrow$ Purchase_Location
- Q4. Customer_ID $\rightarrow$ Store_ID

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The Satisfies Algorithm: Example



Consider the relation TABLE_PURCHASE_DETAIL(Customer_ID, Store_ID, Purchase_Location)

TABLE_PURCHASE_DETAIL

Customer ID	Store ID	Purchase Location
1	1	Los Angeles
1	3	San Francisco
2	1	Los Angeles
3	2	New York
4	3	San Francisco

Check if the following functional dependencies are satisfied in the above relation:

Q1. Customer_ID $\rightarrow$ Purchase_Location

Q2. Store_ID $\rightarrow$ Purchase_Location

Q3. {Customer_ID, Store_ID} $\rightarrow$ Purchase_Location

Q4. Customer_ID $\rightarrow$ Store_ID

Q1. No: Customer_ID = 1, has two different locations

Q2. No: Store_ID = 1 has two purchase locations

Q3. Yes: every combination of Customer ID and Store ID uniquely determines purchase location.

Q4. No: Customer ID = 1, has two different store IDs 1 and 3.

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The Satisfies Algorithm: Example



X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Q. Which of the following functional dependencies are satisfied by the instance?

(a) XY $\rightarrow$ Z and Z $\rightarrow$ Y

(b) YZ $\rightarrow$ X and Y $\rightarrow$ Z

(c) YZ $\rightarrow$ X and X $\rightarrow$ Z

(d) XZ $\rightarrow$ Y and Y $\rightarrow$ X

[GATE 2000]

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The Satisfies Algorithm: Example



X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Q. Which of the following functional dependencies are satisfied by the instance?

- (a) $XY \rightarrow Z$ and $Z \rightarrow Y$
 - (b) $YZ \rightarrow X$ and $Y \rightarrow Z$
 - (c) $YZ \rightarrow X$ and $X \rightarrow Z$
 - (d) $XZ \rightarrow Y$ and $Y \rightarrow X$
- [GATE 2000]

- (a) $XY \rightarrow$ holds, $Z \rightarrow Y$, does not hold.
- (b) $YZ \rightarrow X$ hold, $Y \rightarrow Z$ hold.**
- (c) $YZ \rightarrow X$ hold, $X \rightarrow Z$, does not hold.
- (d) $XZ \rightarrow Y$ does not hold, $Y \rightarrow X$ hold.

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Redundancy in Functional Dependency



- ☐ Given a set F of FDs, a FD $A \rightarrow B$ in F is said to be redundant with respect to the FDs of F if and only if $A \rightarrow B$ is implied and can be derived from a subset F' of F such that $F' \equiv F - \{A \rightarrow B\}$.
- ☐ Eliminating Redundant FDs allows us to minimize the set of FDs.

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The Membership Algorithm



- Used to determine if there exists a redundant FD $A \rightarrow B$ in a given set of functional dependencies F

Input: F and a FD $A \rightarrow B$ belonging to F

Output: TRUE if $A \rightarrow B$ is redundant in F , otherwise FALSE

- **Step 1:** Remove temporarily $A \rightarrow B$ from F . Set $G = F - \{A \rightarrow B\}$. If $G \neq \emptyset$, proceed to Step 2; otherwise halt with output FALSE
- **Step 2:** Initialize the set of attributes T_i with $i=1$ with the set of attribute(s) A , i.e., Set $T_i = T_1 = \{A\}$.
- **Step 3:** Search in G for FDs $X \rightarrow Y$ such that $X \subseteq T_i$.
 - Step 3a: If such FD $X \rightarrow Y$ is found from Step 3, form $T_{i+1} \leftarrow Y \cup T_i$ and assign $i \leftarrow i+1$.
 - Step 3aa: If all the attributes of B belongs to T_i declare the FD $A \rightarrow B$ to be redundant, halt with output TRUE.
 - Step 3ab: If all attributes of B are not members of T_i , assign $G \leftarrow G - \{X \rightarrow Y\}$ and repeat Step 3.
 - Step 3b: If $G = \emptyset$ or there is no such FD is found from Step 3, then halt with output FALSE.

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The Membership Algorithm



Given the set $F = \{X \rightarrow YW, XW \rightarrow Z, Z \rightarrow Y, XY \rightarrow Z\}$. Using membership algorithm, determine if the FD $XY \rightarrow Z$ is redundant in F .

Step#	G	Is $G = \emptyset$	i	T_i	Does $Z \in T_i$?
1	$\{X \rightarrow YW, XW \rightarrow Z, Z \rightarrow Y\}$	No	1	$\{XY\}$	No
2	$\{XW \rightarrow Z, Z \rightarrow Y\}$	No	2	$\{XYW\}$	No
3	$\{Z \rightarrow Y\}$	No	3	$\{XYWZ\}$	Yes

The algorithm halts as $Z \in T_i$ at $i=3$.
 $XY \rightarrow Z$ is redundant in F .

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The Membership Algorithm



Given the set $F = \{X \rightarrow Z, ZW \rightarrow X, Z \rightarrow Y, Z \rightarrow W\}$. Using membership algorithm, determine if the FD $Z \rightarrow Y$ is redundant in F .

Step#	G	Is $G = \phi$	i	T_i	Does $Y \in T_i$?
1	$\{X \rightarrow Z, ZW \rightarrow X, Z \rightarrow Y, Z \rightarrow W\}$	No	1	$\{Z\}$	No
2	$\{X \rightarrow Z, ZW \rightarrow X\}$	No	2	$\{ZW\}$	No
3	$\{X \rightarrow Z\}$	No	3	$\{XZW\}$	No
4	ϕ	Yes	4	$\{XZW\}$	No

The algorithm halts as $G = \phi$ but $Y \notin T_4$.

$Z \rightarrow Y$ is NOT redundant in F .

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Cover



- ☐ G covers F if every FD in F can be inferred from G (i.e., if F^+ is subset of G^+).
- ☐ Two sets of FDs F and G defined over same relation schema are equivalent if F covers G and G covers F
- ☐ G is a non-redundant cover of F if G covers F and no proper subset H of G exist such that $H^+ = G^+$.
- ☐ Canonical Cover/Irreducible Set/ Minimal Cover: Minimum number of FD's required.

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Equivalence of Two FD Sets

❑ Two FD sets F_1 and F_2 are equivalent if

❑ $F_1^+ = F_2^+$

or

❑ F_1 subset of F_2^+ and F_2 is subset of F_1^+

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Equivalence of Two FD Sets

F
 $A \rightarrow C$
 $AC \rightarrow D$
 $E \rightarrow AD$
 $E \rightarrow H$

G
 $A \rightarrow CD$
 $E \rightarrow AH$

Closure of all FDs of F using power of G
 $A^+ \rightarrow ACD$
 $AC^+ \rightarrow ACD$
 $E^+ \rightarrow EAHCD$

Compare

↑

F is subset of G

Closure of all FDs of G using power of F
 $A \rightarrow CD$
 $E \rightarrow AH$

Compare

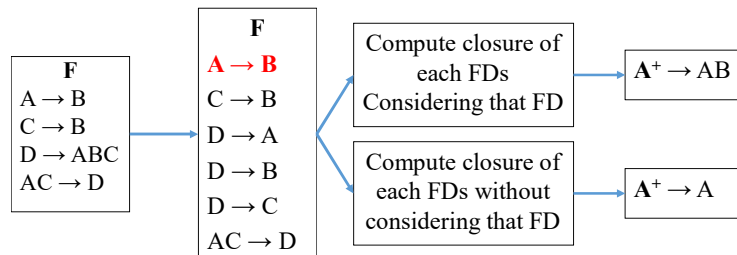
↑

G is subset of F

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

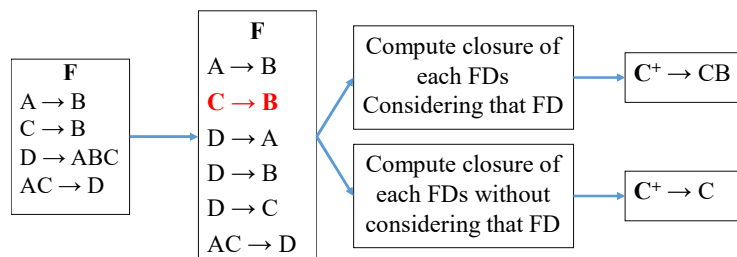


I cant reach to B if I will not use the FD: $A \rightarrow B$, thus it is important (non-redundant)

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

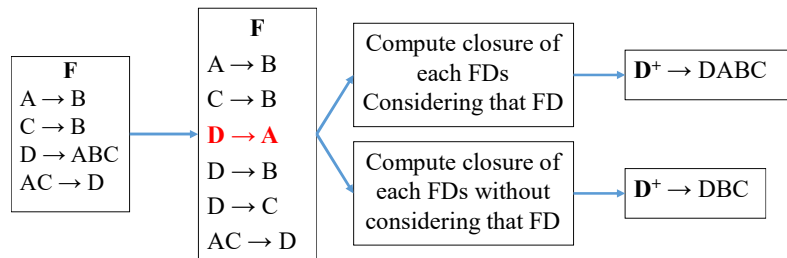


I cant reach to B if I will not use the FD: $C \rightarrow B$, thus it is important (non-redundant)

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

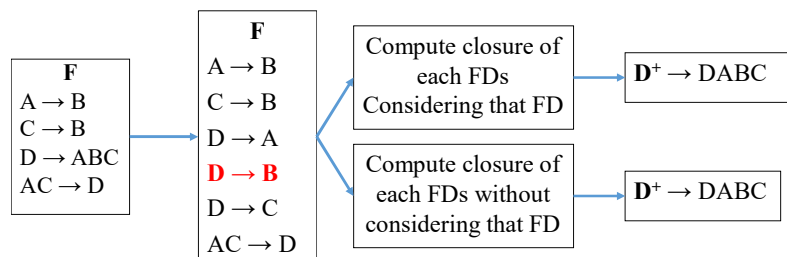


I cant reach to A if I will not use the FD: $D \rightarrow A$, thus it is important (non-redundant)

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

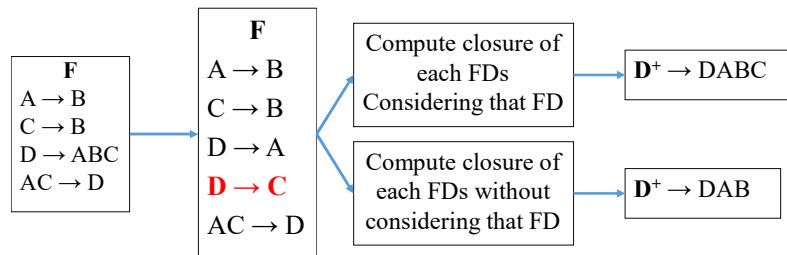


I can get same closure set even if I am not using the FD: $D \rightarrow B$, thus it is not important, remove it from the F

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

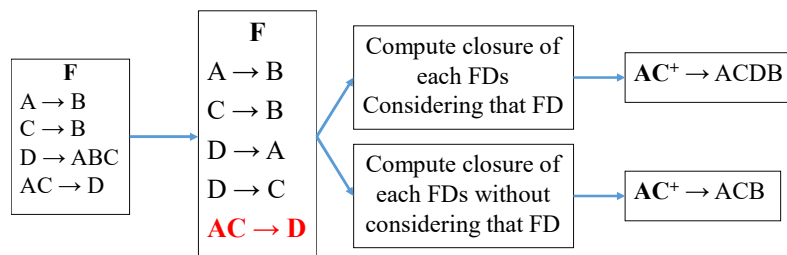


I cant reach to C if I will not use the FD: $D \rightarrow C$, thus it is important (non-redundant)

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

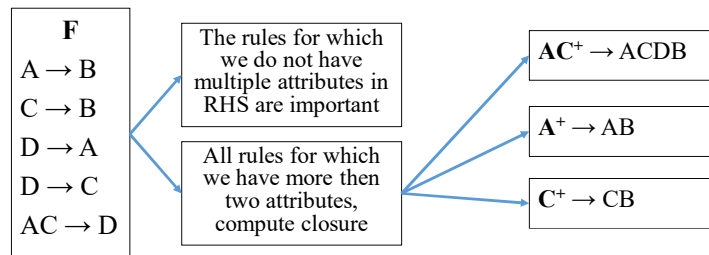


I cant reach to D if I will not use the FD: $AC \rightarrow D$, thus it is important (non-redundant)

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

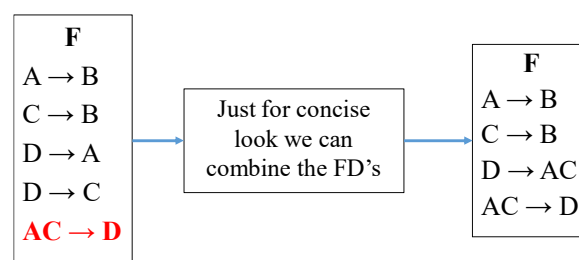


- If I can achieve whatever I am getting for AC^+ only using A^+ then I can say C is redundant in dependency $AC \rightarrow D$.
- If I can achieve whatever I am getting for AC^+ only using C^+ then I can say A is redundant in relation $AC \rightarrow D$.
- Thus writing $AC \rightarrow$ is important.

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Canonical Cover / Irreducible Set

□ **Example:** $R(ABCD)$, $F\{A \rightarrow B, C \rightarrow B, D \rightarrow ABC, AC \rightarrow D\}$

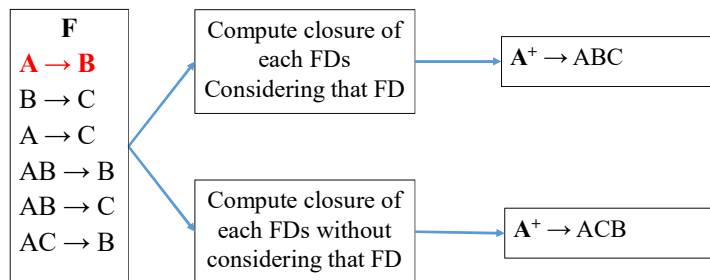


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Canonical Cover / Irreducible Set



□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$



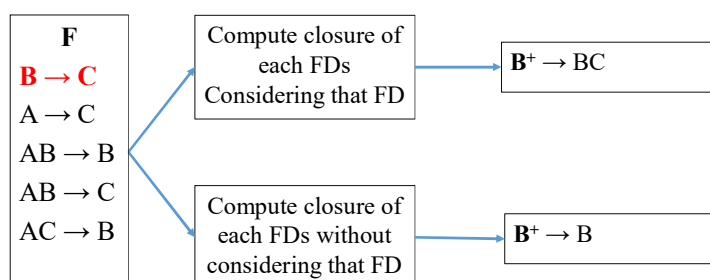
Not Important

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Canonical Cover / Irreducible Set



□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$



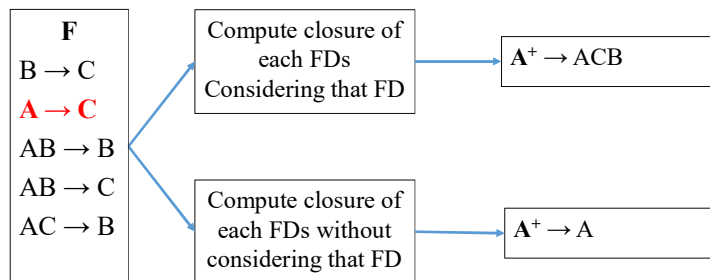
Important

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Canonical Cover / Irreducible Set



□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$



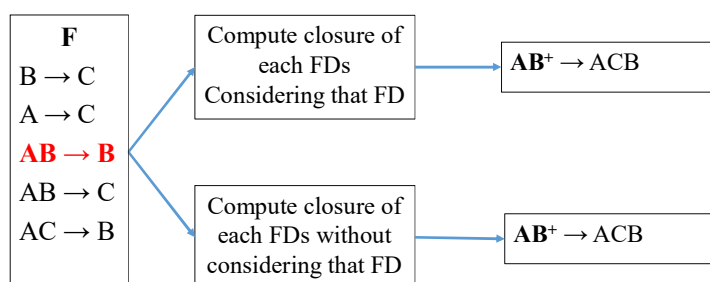
Important

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Canonical Cover / Irreducible Set



□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$



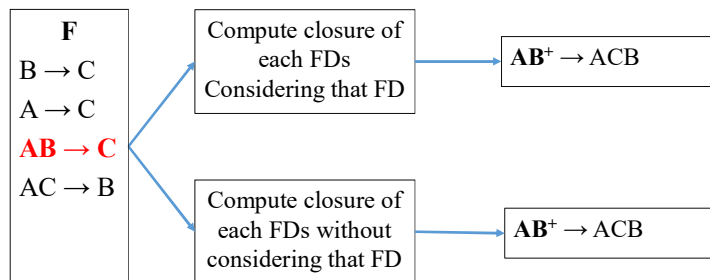
Not Important

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Canonical Cover / Irreducible Set



□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$



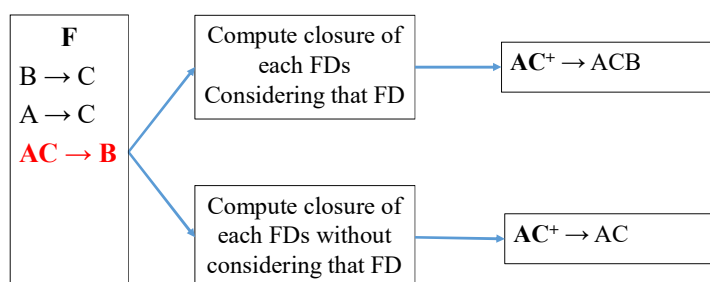
Not Important

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Canonical Cover / Irreducible Set



□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$

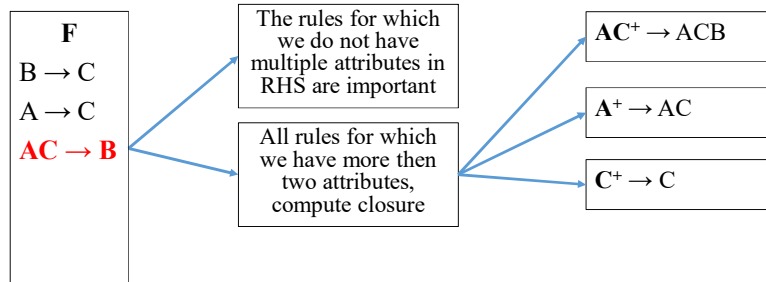


Important

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Canonical Cover / Irreducible Set

□ **Example-2:** $R(ABC)$, $F\{A \rightarrow B, B \rightarrow C, A \rightarrow C, AB \rightarrow B, AB \rightarrow C, AC \rightarrow B\}$



- If I can achieve whatever I am getting for AC^+ only using A^+ then I can say C is redundant in dependency $AC \rightarrow B$.
- If I can achieve whatever I am getting for AC^+ only using C^+ then I can say A is redundant in relation $AC \rightarrow B$.
- Thus dependency $AC \rightarrow B$ is important.

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Keys

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Super Key?



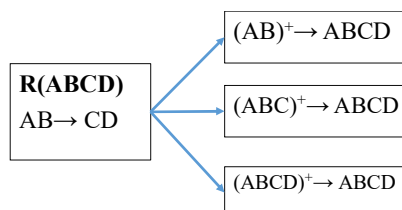
- ❑ A super key is a unique set of attribute(s) that determine the set of other attributes in a relation.
- ❑ In a relation $R(A,B,C,D,E,F)$, we define set of attributes as $P=\{A,B,C,D,E,F\}$. A super key SK is a subset of P ($SK \subseteq P$) that determines all other attributes,
 - i.e., $(SK)^+ = P$
 - There can be many superkeys in a relation (but at least one must exist).
 - A super key is a set of attributes that has the uniqueness property, but is not necessarily minimal.
 - **Candidate key** is a minimal super key, i.e. removing any attribute from a candidate key will not retain its ability to uniquely determine other attributes.
 - Two properties of candidate key, or called just key, are **unique** and **minimal**.

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Super Key: Example



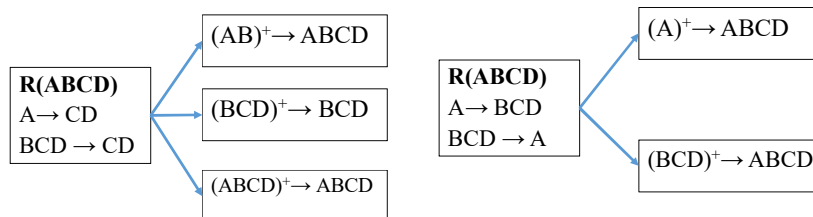
- ❑ Super key can have redundancy.



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Candidate Key

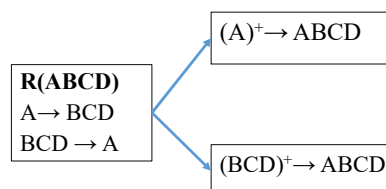
- ❑ A super key is called candidate key if its no proper subset is a super key (also called minimal super key).
- ❑ There should be at least one candidate key.



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Prime Attributes and Primary Key

- ❑ Attribute that are member of at least one candidate keys are called **prime attribute**.
- ❑ If a relation has multiple keys, database designer specifies one of them to be the used as a key while others won't be. This specially selected key is called **primary key**.
- ❑ The candidate keys which do not get elected as primary key are called **alternate keys** or **secondary keys**.



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Surrogate Key



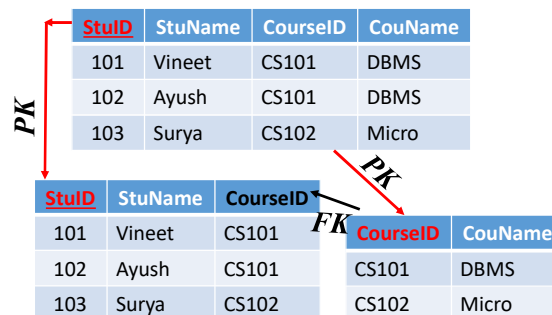
- ❑ Surrogate key, also called a *synthetic primary key*, is generated when a new record is inserted into a table automatically by a database that can be declared as the primary key of that table .
- ❑ It is the sequential number outside of the database that is made available to the user and the application or it acts as an object that is present in the database but is not visible to the user or application.

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Foreign Keys



- ❑ A foreign key is a column or group of columns in a relational database table that refers the primary key of the same table/other table to represent relationship.
- ❑ The concept of referential integrity is derived from foreign key.



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Composite and Secondary Key



- ❑ **Composite:** A key composed of more than one column sometime it is also known as concatenated key.
- ❑ **Secondary:** Key used to speedup search and retrieval contrary to primary key (does not necessary to contain unique value).

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Finding Superkeys/Candidate key



How to find superkeys / candidate keys in a given relation:

Superkeys are those sets of attributes whose closure is the set of all attributes.

Find all superkeys and candidate keys in $F = \{A \rightarrow B, A \rightarrow C, B \rightarrow C\}$

Let us first find the attribute closure of all subsets of the attribute set.

Attribute set	Closure	Closure contains all attributes?	Superkey?
A	$A^+ = \{ABC\}$	Y	Y
B	$B^+ = \{BC\}$	N	N
C	$C^+ = \{C\}$	N	N
AB	$(AB)^+ = \{ABC\}$	Y	Y
BC	$(BC)^+ = \{BC\}$	N	N
AC	$(AC)^+ = \{ABC\}$	Y	Y
ABC	$(ABC)^+ = \{ABC\}$	Y	Y

Superkeys: A; {AB}; {AC}; {ABC}.

Candidate keys are minimal superkeys, i.e., those superkeys, whose proper subsets are not a superkey, are candidate keys.

Candidate keys: A. In this case, there is only one candidate key!

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Gate Questions



Q. An instance of relational schema $R(A, B, C)$ has distinct values for attribute A. Can you conclude that A is a candidate key for R?
[GATE1994]

- $A \rightarrow BC$ holds, $A^+ \{A, B, C\}$ for the given instance.
- FD values are defined on the schema and not on the instance, thus based on the instance we cannot say, what hold for schema.
- Candidate key is defined over schema and not instance.

A	B	C
1	5	6
2	4	7
3	4	5

Instance 1 of the R

A	B	C
1	3	6
2	4	7
3	4	5

Instance 2 of the R

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Gate Questions



Q. Consider a relation scheme $R(A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R?
(a) AE, BE
(b) AE, BE, DE
(c) AEH, BEH, BCH
(d) AEH, BEH, DEH
[GATE2005]

- $A^+ \{A, B\}$
- $E^+ \{E, C\}$
- $D^+ \{D, A, B\}$
- $AE^+ \{ABCDE\}$
- $BE^+ \{ABCDE\}$
- $DE^+ \{ABCDE\}$
- AEH, BEH, DEH {add H in previous three}.

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Gate Questions



Q. A Relation R with FD set $\{A \rightarrow BC, B \rightarrow A, A \rightarrow C, A \rightarrow D, D \rightarrow A\}$. How many candidate keys will be there in R?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

- A+: {A, B, C, D}
- B+: {A, B, C, D}
- D+: {A, B, C, D}

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Gate Questions



Q. Relation R has eight attributes A,B,C,D,E,F,G,H. Fields of R contain only atomic values. $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?

- (a) 3
- (b) 4
- (c) 5
- (d) 6

[GATE2013]

- To derive all attributes, at least D must be included because no dependency given for D.
- D,A,F = {ABCD FHEG}
- DBF+ = {ABCD FHEG}
- DEB = {ABCD FHEG}
- DECH = {ABCD FHEG}

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Gate Questions



Q. The maximum number of superkeys for the relation schema $R(E, F, G, H)$ with E as the key is:

- (a) 5
- (b) 6
- (c) 7
- (d) 8

[GATE2014]

- Any set which include E will be a super key (because E is key).
- Rest (F, G, H) = super set for this = $2^3=8$
- Superkeys will be
- E
- EF
- EG
- EH
- EFG
- EFH
- EGH
- $EFGH$

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Gate Questions



Q. Which of the following is NOT a superkey in a relational schema with attributes V, W, X, Y, Z and primary key VY ?

- (a) $VXYZ$
- (b) $VWXZ$
- (c) $VWXY$
- (d) $VWXYZ$

[GATE2016]

- If VY is not present then that will not be a superkeys (because superkeys must include key).
- Option B does not Y thus it is not a superkey.

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Gate Questions (Solve this)



Q. Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M\}$ and the set of functional dependencies $\{ \{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, \{K\} \rightarrow \{M\}, \{L\} \rightarrow \{N\} \}$ on R . What is the key for R ?

- (a) $\{E, F\}$
- (b) $\{E, F, H\}$
- (c) $\{E, F, H, K, L\}$
- (d) $\{E\}$

[GATE2014]

Q. The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:
 $V \rightarrow W$; $VW \rightarrow X$; $Y \rightarrow VX$; $Y \rightarrow Z$

Which of the following is irreducible equivalent for this set of functional dependencies?

- (a) $\{V \rightarrow W; V \rightarrow X; Y \rightarrow V; Y \rightarrow Z\}$
- (b) $\{V \rightarrow W; W \rightarrow X; Y \rightarrow V; Y \rightarrow Z\}$
- (c) $\{V \rightarrow W; V \rightarrow X; Y \rightarrow V; Y \rightarrow X; Y \rightarrow Z\}$
- (d) $\{V \rightarrow W; W \rightarrow X; Y \rightarrow V; Y \rightarrow X; Y \rightarrow Z\}$

[GATE2017]